



# Python Libraries for Cybersecurity

*(Introductory → Applied → Advanced Toolkit Resource Guide)*

This curated resource guide is designed for entry-level to early-intermediate cybersecurity learners () and aligns with foundational workforce competencies reflected in the **NICE Workforce Framework (NIST SP 800-181 Rev.1)** published by the National Institute of Standards and Technology.

The libraries below are categorized by function and skill level to support structured learning pathways and project-based application.



## Foundational Python Libraries (Beginner-Friendly)

These are essential for building cybersecurity scripts and understanding core concepts.



### hashlib

**Purpose:** Hashing passwords and files

**Use Cases:** SHA-256, MD5 (education only), file integrity checks

```
import hashlib

password = "securepassword"
hash_object = hashlib.sha256(password.encode())
print(hash_object.hexdigest())
```



Docs: <https://docs.python.org/3/library/hashlib.html>

NICE Alignment: Encryption & data integrity knowledge (K0157)



### secrets

**Purpose:** Secure random token generation

**Use Cases:** Password generators, session tokens

```
import secrets
print(secrets.token_hex(16))
```

🔗 Docs: <https://docs.python.org/3/library/secrets.html>  
NICE Alignment: Secure authentication principles



**Purpose:** Basic network communication  
**Use Cases:** Simple port scanning (educational), network basics

🔗 Docs: <https://docs.python.org/3/library/socket.html>  
NICE Alignment: Networking fundamentals (K0001)

## 2 Applied Cybersecurity Libraries



**Purpose:** Packet crafting and network analysis  
**Use Cases:** Traffic inspection, packet sniffing (lab use only)

```
from scapy.all import sniff
packets = sniff(count=5)
packets.summary()
```

🔗 <https://scapy.net/>  
NICE Alignment: Cyber Defense Analysis exposure



**Purpose:** Advanced encryption & key management  
**Use Cases:** AES encryption, secure key handling

🔗 <https://cryptography.io/>  
NICE Alignment: Applied cryptographic principles



**Purpose:** Web request automation

**Use Cases:** API security testing basics, header inspection

```
import requests
response = requests.get("https://example.com")
print(response.headers)
```

 <https://docs.python-requests.org/>

NICE Alignment: Web application awareness

## **Defensive & Monitoring Libraries**

 **logging**

**Purpose:** System logging

**Use Cases:** Building secure applications with traceability

 <https://docs.python.org/3/library/logging.html>

NICE Alignment: Log analysis fundamentals (S0078)

 **os & pathlib**

**Purpose:** File system monitoring

**Use Cases:** File integrity checks, scanning directories

 **yara-python**

**Purpose:** Malware pattern matching

**Use Cases:** Signature-based detection

 <https://yara.readthedocs.io/>

NICE Alignment: Malware analysis exposure

## Ethical Hacking & Testing (Educational Use Only)

 `python-nmap`

**Purpose:** Network scanning wrapper

**Use Cases:** Port scanning labs

 <https://xael.org/pages/python-nmap-en.html>

 Must be used only in controlled lab environments.

 `pwntools`

**Purpose:** Exploit development education

**Use Case:** CTF competitions

## AI & Cybersecurity Integration

 `scikit-learn`

**Purpose:** Anomaly detection

**Use Cases:** Intrusion detection prototypes

 <https://scikit-learn.org/>

 `pandas`

**Purpose:** Log analysis

**Use Cases:** Filtering suspicious events

```
import pandas as pd
logs = pd.read_csv("logs.csv")
print(logs.head())
```

## **transformers (Hugging Face)**

**Purpose:** NLP-based phishing detection

**Use Case:** Email classification

 <https://huggingface.co/docs/transformers>

## **Suggested Learning Progression**

| Level   | Focus              | Libraries                  |
|---------|--------------------|----------------------------|
| Level 1 | Security basics    | hashlib, secrets, logging  |
| Level 2 | Network awareness  | socket, requests           |
| Level 3 | Packet inspection  | scapy                      |
| Level 4 | Applied encryption | cryptography               |
| Level 5 | Log analytics      | pandas                     |
| Level 6 | AI detection       | scikit-learn, transformers |

## **NICE/NIST Workforce Framework Alignment**

This resource supports exposure to the following NICE work roles:

- Cyber Defense Analyst (PR-CDA-001)
- Security Operations Support
- Entry-Level Security Practitioner

Primary Reference:

NIST SP 800-181 Rev.1 – NICE Workforce Framework

<https://csrc.nist.gov/publications/detail/sp/800-181/rev-1/final>

## **Suggested Mini-Projects Using These Libraries**

1. Password strength checker (hashlib + secrets)
2. Log anomaly detection (pandas + scikit-learn)
3. Packet inspection demo (scapy)
4. File integrity monitor (hashlib + os)
5. Phishing email classifier (transformers)